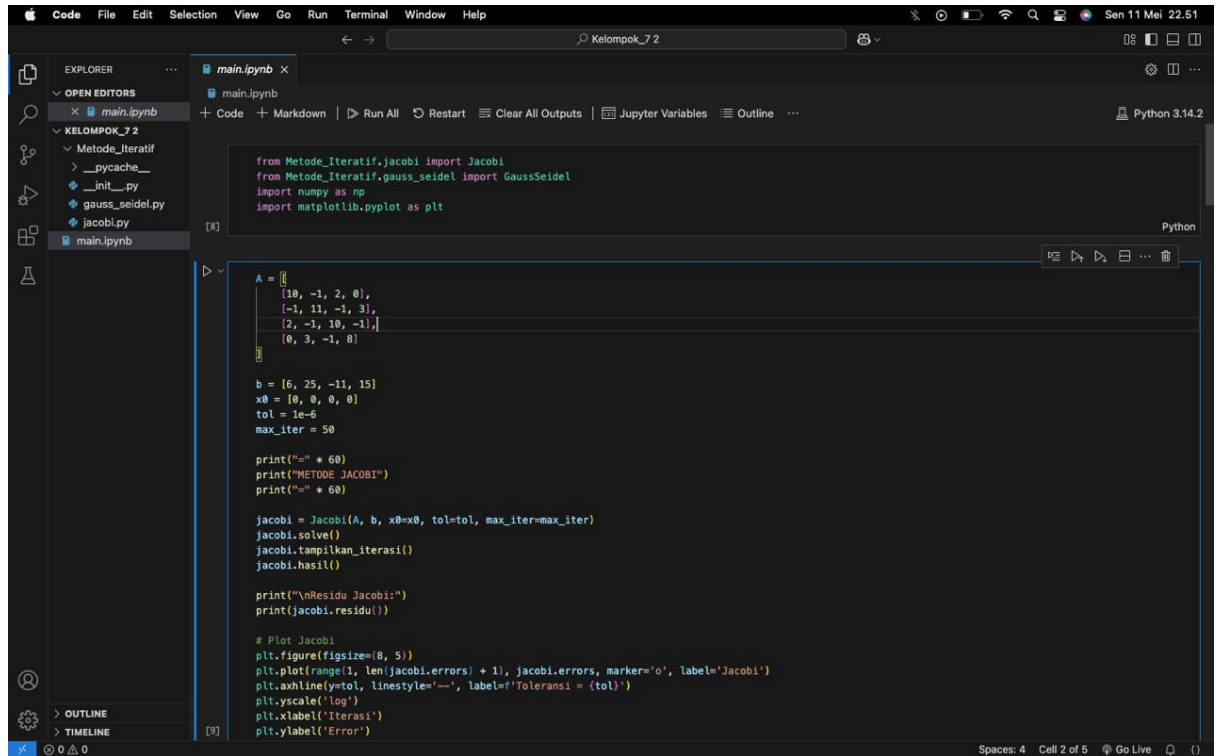


Kelompok 7

Fitri Rachmadani 230101501030
Fatur Rahim 230101501029
Nabila Salsabil 230101501025

- Contoh penggunaan program Hasil eksekusi dan analisis Plot/visualisasi grafik



```
from Metode_Iteratif.jacobi import Jacobi
from Metode_Iteratif.gauss_seidel import GaussSeidel
import numpy as np
import matplotlib.pyplot as plt

A = [
    [10, -1, 2, 0],
    [-1, 11, -1, 3],
    [2, -1, 10, -1],
    [0, 3, -1, 8]
]

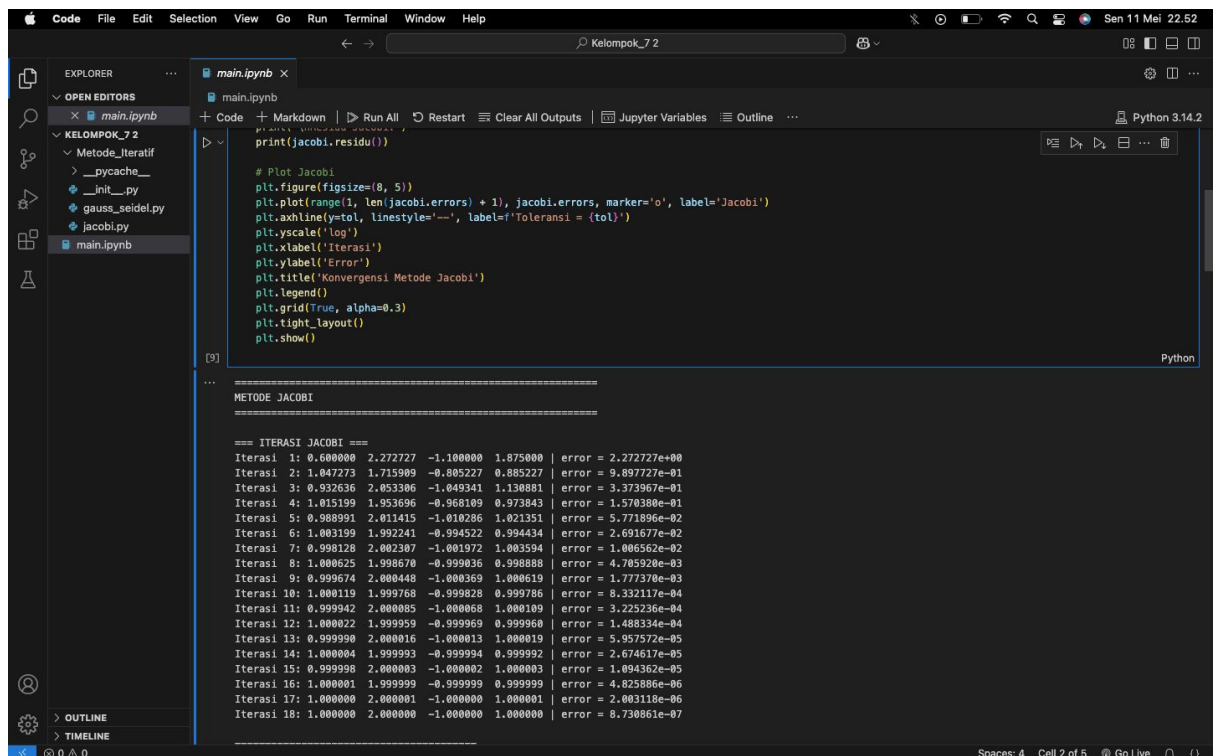
b = [6, 25, -11, 15]
x0 = [0, 0, 0, 0]
tol = 1e-6
max_iter = 50

print("=" * 60)
print("METODE JACOBI")
print("=" * 60)

jacobi = Jacobi(A, b, x0=x0, tol=tol, max_iter=max_iter)
jacobi.solve()
jacobi.tampilkan_iterasi()
jacobi.hasil()

print("\nResidu Jacobi:")
print(jacobi.residu())

# Plot Jacobi
plt.figure(figsize=(8, 5))
plt.plot(range(1, len(jacobi.errors) + 1), jacobi.errors, marker='o', label='Jacobi')
plt.axhline(y=tol, linestyle='--', label='Toleransi = {tol}')
plt.yscale('log')
plt.xlabel('Iterasi')
plt.ylabel('Error')
```

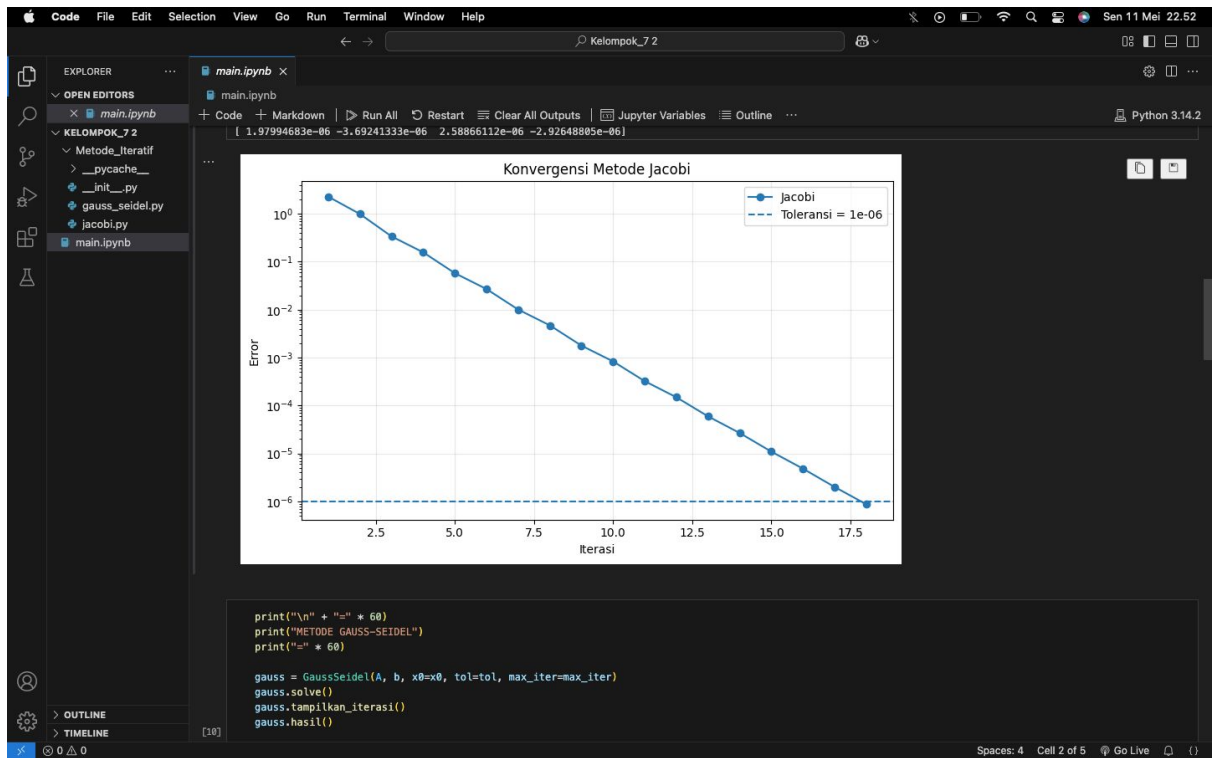


```
print(jacobi.residu())

# Plot Jacobi
plt.figure(figsize=(8, 5))
plt.plot(range(1, len(jacobi.errors) + 1), jacobi.errors, marker='o', label='Jacobi')
plt.axhline(y=tol, linestyle='--', label='Toleransi = {tol}')
plt.yscale('log')
plt.xlabel('Iterasi')
plt.ylabel('Error')
plt.title('Konvergensi Metode Jacobi')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

=====
METODE JACOBI
=====

=== ITERASI JACOBI ===
Iterasi 1: 0.600000 2.272727 -1.100000 1.875000 | error = 2.272727e+00
Iterasi 2: 1.047273 1.715909 -0.805227 0.805227 | error = 9.897727e-01
Iterasi 3: 0.932636 2.053306 -1.049341 1.130881 | error = 3.373367e-01
Iterasi 4: 1.015199 1.853696 -0.969109 0.973843 | error = 1.570300e-01
Iterasi 5: 0.988991 2.011415 -1.010286 1.021351 | error = 5.771896e-02
Iterasi 6: 1.003199 1.992241 -0.994522 0.994434 | error = 2.691677e-02
Iterasi 7: 0.998128 2.002307 -1.001972 1.003594 | error = 1.006562e-02
Iterasi 8: 1.000625 1.998670 -0.999036 0.998888 | error = 4.705920e-03
Iterasi 9: 0.999674 2.000448 -1.000369 1.000619 | error = 1.777370e-03
Iterasi 10: 1.000119 1.999768 -0.999828 0.999786 | error = 8.332117e-04
Iterasi 11: 0.999942 2.000085 -1.000068 1.000109 | error = 3.225236e-04
Iterasi 12: 1.000022 1.999959 -0.999969 0.999960 | error = 1.488334e-04
Iterasi 13: 0.999990 2.000016 -1.000013 1.000019 | error = 5.957572e-05
Iterasi 14: 1.000004 1.999993 -0.999994 0.999992 | error = 2.674617e-05
Iterasi 15: 0.999998 2.000003 -1.000002 1.000003 | error = 1.094362e-05
Iterasi 16: 1.000001 1.999999 -0.999999 0.999999 | error = 4.825886e-06
Iterasi 17: 1.000000 2.000001 -1.000000 1.000001 | error = 2.003118e-06
Iterasi 18: 1.000000 2.000000 -1.000000 1.000000 | error = 8.730861e-07
```



Code File Edit Selection View Go Run Terminal Window Help

Kelompok_7_2

main.ipynb

main.ipynb

KELOMPOK_7_2

Metode Iteratif

__pycache__

__init__.py

gauss_seidel.py

jacobi.py

main.ipynb

gauss.tampilkan_iterasi()
gauss.hasil()

print("\nResidu Gauss-Seidel:")
print(gauss.residu())

Python

=====

METODE GAUSS-SEIDEL

=====

== ITERASI GAUSS-SEIDEL ==

Iterasi	x1	x2	x3	x4	error
1	0.600000	2.327273	-0.987273	0.878864	error = 2.327273e+00
2	1.030182	2.036938	-1.014456	0.984341	error = 4.301818e-01
3	1.006585	2.003555	-1.002527	0.998351	error = 3.338300e-02
4	1.000861	2.000298	-1.000307	0.999850	error = 5.724063e-03
5	1.000091	2.000021	-1.000031	0.999988	error = 7.696983e-04
6	1.000008	2.000001	-1.000003	0.999999	error = 8.291662e-05
7	1.000001	2.000000	-1.000000	1.000000	error = 7.697313e-06
8	1.000000	2.000000	-1.000000	1.000000	error = 6.220629e-07

=====

HASIL METODE GAUSS-SEIDEL

=====

x1 = 1.000000004
x2 = 1.999999999
x3 = -1.000000001
x4 = 1.000000000
Jumlah iterasi : 8
Error akhir : 6.22062942e-07
Status : Konvergen

=====

Residu Gauss-Seidel:

[4.22335984e-07 -8.89315714e-08 -3.57543506e-08 -1.77635684e-15]

print("\n" + "=" * 60)
print("PERBANDINGAN LENGKAP JACOBI VS GAUSS-SEIDEL")

Spaces: 4 Cell 2 of 5 Go Live

```
Code File Edit Selection View Go Run Terminal Window Help
Kelompok_7_2

EXPLORER
main.ipynb
main.ipynb
KELOMPOK_7_2
Metode_Iteratif
  __pycache__
  __init__.py
  gauss_seidel.py
  jacobi.py
main.ipynb

main.ipynb
+ Code + Markdown | ▶ Run All | ⌂ Restart | Clear All Outputs | Jupyter Variables | Outline ...
Python 3.14.2

print("\n" + "=" * 60)
print("PERBANDINGAN LENGKAP JACOBI VS GAUSS-SEIDEL")
print("=" * 60)

solusi_exact = np.linalg.solve(np.array(A, dtype=float), np.array(b, dtype=float))

print("\nSolusi Eksak (NumPy):")
for i, val in enumerate(solusi_exact, start=1):
    print(f"x({i}) = {val:.8f}")

print("\nHasil Akhir Jacobi:")
for i, val in enumerate(jacobi.x, start=1):
    print(f"x({i}) = {val:.8f}")

print("\nHasil Akhir Gauss-Seidel:")
for i, val in enumerate(gauss.x, start=1):
    print(f"x({i}) = {val:.8f}")

print("\nJumlah Iterasi:")
print(f"Jacobi      : {jacobi.iterations}")
print(f"Gauss-Seidel : {gauss.iterations}")

print("\nError Akhir:")
print(f"Jacobi      : {jacobi.errors[-1]:.8e}")
print(f"Gauss-Seidel : {gauss.errors[-1]:.8e}")

print("\nResidu:")
print(f"Jacobi      : {jacobi.residu():.8e}")
print(f"Gauss-Seidel : {gauss.residu():.8e}")

if jacobi.iterations > gauss.iterations:
    print("\nKesimpulan: Gauss-Seidel lebih cepat konvergen dibanding Jacobi.")
elif jacobi.iterations < gauss.iterations:
    print("\nKesimpulan: Jacobi lebih cepat konvergen dibanding Gauss-Seidel.")
else:
    print("\nKesimpulan: Keduanya memiliki jumlah iterasi yang sama.")

[11] Python
Spaces: 4 Cell 2 of 5 Go Live
```

```
Code File Edit Selection View Go Run Terminal Window Help
Kelompok_7_2

EXPLORER
main.ipynb
main.ipynb
KELOMPOK_7_2
Metode_Iteratif
  __pycache__
  __init__.py
  gauss_seidel.py
  jacobi.py
main.ipynb

main.ipynb
+ Code + Markdown | ▶ Run All | ⌂ Restart | Clear All Outputs | Jupyter Variables | Outline ...
Python 3.14.2

=====
PERBANDINGAN LENGKAP JACOBI VS GAUSS-SEIDEL
=====

Solusi Eksak (NumPy):
x1 = 1.00000000
x2 = 2.00000000
x3 = -1.00000000
x4 = 1.00000000

Hasil Akhir Jacobi:
x1 = 1.00000014
x2 = 1.99999976
x3 = -0.99999982
x4 = 0.99999975

Hasil Akhir Gauss-Seidel:
x1 = 1.00000004
x2 = 1.99999999
x3 = -1.00000001
x4 = 1.00000000

Jumlah Iterasi:
Jacobi      : 18
...
Jacobi      : [ 1.97994683e-06 -3.69241333e-06  2.58866112e-06 -2.92648805e-06]
Gauss-Seidel : [ 4.22335984e-07 -8.89315714e-08 -3.57543506e-08 -1.77635684e-15]

Kesimpulan: Gauss-Seidel lebih cepat konvergen dibanding Jacobi.
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

# Plot perbandingan
plt.figure(figsize=(8, 5))
plt.plot(range(1, len(jacobi.errors) + 1), jacobi.errors, marker='o', label='Jacobi')
plt.plot(range(1, len(gauss.errors) + 1), gauss.errors, marker='s', label='Gauss-Seidel')
plt.axhline(y=tol, linestyle='--', label=f'Toleransi = {tol}')
plt.yscale('log')
plt.xlabel('Iterasi')
```

